INCH-POUND

A-A-58073
31 JULY 1996

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COMMERCIAL ITEM DESCRIPTION

TEST SET, ULTRASONIC

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This Commercial Item Description (CID) covers the minimum performance and technical requirements for a portable test set,ultrasonic for nondestructive flaw testing and thickness gauging of metallic and nonmetallic materials.

2. SALIENT CHARACTERISTICS.

2.1 General. The test set, ultrasonic shall provide field and depot level nondestructive inspection technicians with a rugged, portable and reliable means for performing ultrasonic inspections for crack detection, corrosion testing, and thickness testing. The test set shall be capable of performing pulse-echo, through-transmission, and pitch-catch modes of ultrasonic inspections using typical ultrasonic transducers found in the Air Force inventory. Particular emphasis shall be given to design features that maximize reliability, maintainability, and ease of use.

Beneficial comments(recommendations, additions, deletions) and any pertinent data which may be of use in improving the document should be addressed to San Antonio ALC/TILDD, 306 Tinker Drive, Bldg. 207, Kelly AFB TX 78241-5916 Document Improvement Proposal(DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6635

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2.2 Gain. Adjustable in steps of 110dB in steps of 0.5dB, with a fine gain of 0.05dB steps.

2.2.1 Gain steps selectable between 0.5/1.0/2.0/6.0 and programmable from 6.0dB to 12dB minimum in 0.5dB steps.

2.3 Pulser. A square wave pulser with a mininum 150 V to maximum 300 V peak pulse into a 100 ohm load or variable pulse width between 30 and 1000 nanoseconds in 10 nanoseconds steps.

2.3.1 PulseWidth. The pulser shall generate the required amplitude pulse over an adjustable range of at least 30 nS minimum to 1000 nS maximum as measured at the half power (50% amplitude) points. The test set shall display the value of this parameter within 10% of its actual value or 10 nS whichever is larger. Step size requirements are 10 nS minimum.

2.3.2 Pulse Reptition Frequency. The pulse repetition frequency from 50 to 4000 Hz automatically coupled to the display and gates or freely adjustable.

2.3.3 Modes of Operation. The pulser modes of operation are pulse-echo, through-transmission, and pitch-catch which are selectable by switch on the panel.

2.4 Receiver.

2.4.1 Receiver Noise. The receiver noise is not to exceed 15 uV RMS (60% grass level on screen at maximum gain in wideband) and shall not exceed 15 uV in any operating band referenced to the input.

2.4.2 Receiver Sensitivity. The receiver shall be capable of responding to a 200 uV peak to peak input signal with full screen indication. This response shall have a minimum 10 dB signal to noise ratio in the wide band operating mode.

2.5 Display. The display shall be a-scan type display with signal freeze, peak hold and display average of the last 2-16 frames. Stored frames and active frame can be displayed simultaneously for comparative purpose.

2.5.1 The display screen shall be a minimum of 12 square inches, 4400 pixels and shall have Video Data Unit(VDU) output of 60 Hz NTSC..

2.5.2 The display start range shall be from -1 inch to 200 inches of aluminum.

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2.5.3 The test set display shall provide a one-to-one correlation between the signal displayed and the actual ultrasonic test signal. The display rate may be less than the inspection repetition rate provided the complete ultrasonic signal is generated and processed, and the flaw alarm operates at the specified inspection rate and not at the display rate.

2.6. Tuning and Filtering. The receiver shall have narrow band filter selections with center frequencies at 0.5, 1, 2.25, 5, 10, and 15 MHz ±10%. It shall have wide band filter selection with a 3 dB operating response within 0.5 - 20 MHz and a high pass filter selection within 3-20 MHz at 3 dB.

2.8 Gates. Two independent gates which one acts as a interface start gate, are to be adjustable from 10%to 90% of the full screen height.

2.9 Measurement. The resolution of the thickness readings shall be a maximum 0.001 inch with precision to 0.0005 in. Thickness readings selectable as initial pulse(IP) to gate 1; IP to gate 2; Interface gate to gate 2; gate 1 to gate 2.

2.9.1 Measurement Accuracy. The minimum thickness measurement accuracy shall be ±0.0015" in aluminum within the range of 0.010 - 0.2", ±0.5% from 1.0" - 5.0", with a dedicated thickness measurement transducer that can reproduce this accuracy in the field. Three thickness measurement transducers of this type shall be provided. These transducers shall not be dependent on serialization to a specific test set to meet the accuracy requirements specified herein.

2.10 Power requirements. The power requirement are that the test set shall operate from 120 VAC to 230 VAC which shall be autoselect.

2.10.1 The test set shall be capable of operating off battery of the pack for at least 8 hours at 78ºF (26ºC) with an indicator on the front of the test set for warning of low battery power for minimum of 30 minutes. The battery pack shall be provide power for 4 hours of operation at 0ºF (-18ºC) before recharging.

2.10.2 The battery charger shall be capable of charging the battery pack while the test set is in operation. The battery charger can be either an integral part of the test set or a separate module.

2.10.3 The test test shall be provided with two battery packs.

2.11 The test set shall have the RS-232C interface with cable and handshake software.

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2.12 Test set Support. The test set shall be equipped with high impact supports (e.g., feet, legs, pads, rails, etc.) on the bottom surface normally in direct contact with the supporting surface. The supports shall be fabricated from a material that will not scratch or mar aircraft surfaces yet will provide sufficient friction to prevent the test set from slipping off of a bare aluminum, as rolled, surface tilted 20° from horizontal. The supports shall be securely fastened to the test set so as to preclude loosening or detachment during field operations.

2.13 Controls. All controls used to adjust the inspection parameters shall be clearly and legibly marked with regard to function and values of adjustment and must be operable with gloved hands.

2.14 Dimensions. The test set shall be a maximum 10 inches high, 15 inches in width, 15 inches deep. The weight of the test set shall be a maximum of 30 pounds with battery pack.

2.15 The test set shall be capable of operating in hazardous explosive area which shall have the been approved by the Underwriter's Laboratories (UL), Factory Mutual (FM) or reputable independent testing laboratory.

3. REGULATORY REQUIREMENTS.

3.1 Recoverable materials. The contractor is encouraged to use recovered materials to the maximum extent practical, in a accordance with paragraph 23.403 of the Federal Acquisition Regulation(FAR)..

3.2 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pounds test set, provided they fall within specified tolerances using conversion tables contained in the latest revision of Federal Standard No. 376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed tolerances specified in the inch-pound test sets, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

4. QUALITY ASSURANCE PROVISIONS.

4.1. Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the proudcer's own drawings, specification, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The goverment reserves the right to require proof of such conformance.

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4.2 Market Acceptance. The following market acceptance criteria are necessary to document the quality, reliability and supportability of the product to be provided under this CID.

4.2.1 The company producing the item must have been producing a test set, ultrasonic meeting the requirements of this CID for at least 2 years. The company must have sold at least 100 test sets of the model.

5. Preservation, packaging, packing, labeling, and marking. Preservation, packaging, packing, labeling, and marking shall be specified in the contract or order.

6.0 NOTES

6.2 Procurement data. The procuring activity should specify the preferred options permitted herein and include the following information in procurement documents:

a) Title, number, and date of this description.
b) Specify special packaging requirements.
c) certification of commercial item prior to first contract delivery.

6.3 National Stock Number(s).

NSN	REFERENCE NO.	CAGEC
6635-01-417-5467	USD15/AF	52039

6.4 Sources of documents.

6.4.1 Federal Standards may be obtained from the General Services Administration, Federal Supply Service, Office of Commodity Management, Washington, DC, 20406.

6.4.2 Federal Acquisition Regulations may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC, 20402.

6.4.3 Electronic Industries Association(EIA) document may be obtained from EIA, Administration Dept, 2001 I ST N W, Washington, DC, 20006-1807.

MILITARY INTEREST:

CIVIL AGENCY COORDINATING ACTIVITY: GSA-FSS

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CUSTODIANS:
Air Force-99

PREPARING ACTIVITY:
AIR FORCE-82

Review Activities:
DLA - GS

PROJECT NUMBER: 6635-0155

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waiver any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	A-A-58073	96/07/31

3. DOCUMENT TITLE
TEST SET, ULTRASONIC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	e. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE *(Include Area Code)*	
SA-ALC/TILDD	(1) Commercial (210) 925-6314	(2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* 306 TINKER DR, BLD 207 KELLY AFB TX 78241-5916	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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